SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 30%

Duration : 45 Minutes

QUESTIONS: 2 **Total Marks:20**

Annexure A

Descriptive Written Test

Code of Conduct:

I Nithya shree . K [192521145] (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Nithya shree

Descriptive Written Test

The six uninformed search algorithms

1. Breadth-First Search (BFS)
2. Uniform-Cost Search (UCS)
3. Depth-First Search (DFS)
4. Depth-Limited Search (DLS)
5. Iterative Deepening Search (IDS)
6. Bidirectional Searc

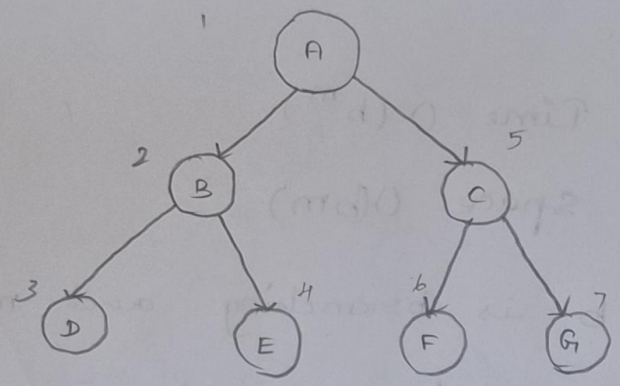
—	~~25~~ 9/10/26 Artifical Intelligence	Nithya shree. K
09/10/26	CO1 - Assesment - 03	192501145
		CSA1712

01> DFS (Depth first search)

A) Definition:

Depth first search (DFS) is an uninformed search algorithm that explores the deepest node of a search tree and ~~back~~ track to other node after reaching the depest node.

Diagram:



Working:

A → B → D → E → C → F → G

Algorithm:

* Starts from root node
* Search/explores till it visits the unvisited child node
* back track to other node afte the depth
* Continue it till it reachs the goal.

Advantages:

less memory using, and simple implementation.

disadvantages:

chances to get stuck in loop, Can't assure shortest path. implementation

Complexity:

Time $O(b^m)$

space $O(bm)$

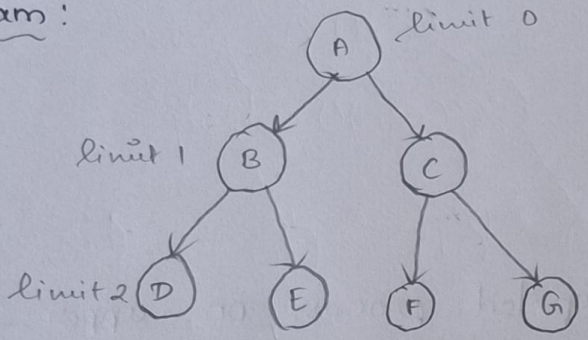
where b is branching and m is maxim depth of a free. search.

02> IDS (iterative deepening search)

1) Definition:

Iterative deepening search (IDS) is an uninformed search algorithm that performes by combaining depth first search (DFS) with Complete Breadth first search (BFS) repeatedly Performes with increasing the limit. to reach the goal.

Diagram:



Working:

Limit 0 — A

Limit 1 — A → B → C

Limit 2 — A → B → C → D → E

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Algorithm:

* Starts from root node
* Search for goal if found stops the search
* if not it increase the limit by 1
* it repeats the process by increasing limit
* it it reaches the goal

Advantages:

complete Branching factors with maximum depth,

disadvantages:

* Repeted process on upper-level nodes
* again and again searching same nod to the shallowest shell.

Complexity:

Time $O(b^d)$

Space $O(bd)$

where d is the shallowest shell.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand